

Personal Information

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Profile

I started my career as a system engineer back in 2008, and over the years built up deep fullstack engineering experience across Python, TypeScript and cloud platforms. That gives me hands-on breadth and enables me to quickly get up to speed with any kind of challenge.

As a Dutch citizen and fluent speaker of English, Dutch and Serbian, I have worked extensively with Dutch 🇳🇱, American 🇺🇸 and British 🇬🇧 customers.

I design and ship LLM-based applications, evals infrastructure, and realtime AI services. It has started with Amelia, a first classifier-based implementation of agentic workflow up to Delphyr, where we are building a custom LLM harness and the integrations with third parties.

You can browse some of my experiments on [GitHub](#), including [agentic usage with MCP servers](#).

I love to spend my free time with family, to explore emerging technologies and take care of my health via balanced diet and strength training. I became fluent in Dutch a long time ago, and I love to keep it brushed up by using it daily at work.

And as a Dutchman, I take my kids to school in our Urban Arrow bakfiets (cargobike). So fun and "gezellig" 🎉 (joyful in Dutch)

Professional experience

From 07/2025 - now **Lead AI Engineer at Delphyr B.V.**

- I joined Delphyr to build an AI platform specifically for the medical field. We are bringing AI into real clinical workflows, from hospitals to GP practices.
- We won our first contract with a Dutch mental health provider, by successfully completing integration with an EHR. On top of that I lead EHR integrations across four vendors
- AI-native engineering is the core of my workflow, with several coding agents in parallel, each running on a plan from Linear and reviewed by a second model
- Python, FastAPI, Pydantic AI, PostgreSQL, Next.js, TypeScript, Tailwind, Auth0, Pinecone, LangWatch, Sentry, Grafana, Docker, Kubernetes, ArgoCD, Nebius and Azure have been used to build the solution
- I led the effort to encrypt data with KEK/DEK envelope encryption and to iterate on pentests to harden the system. I automated the UDI-PI release process
- With the product platform in place, I now focus on the AI layer. This consists of prompt design for various workflows, guardrails, evals, integrations and the observability that keeps model behavior measurable in production
- I built release gating on measured eval regressions and cut the eval suite runtime from 55 to 25 minutes. LangWatch observability with PHI-safe tracing shows where the chat harness needs improvement
- I ran embedding migration from Azure to Nebius, first evaluating and then implementing the migration. I migrated inference providers (Nebius, Baseten) and enabled visibility into token spending

- I worked on prompt design and grounding for ambient clinical documentation, from per-specialism templates to transcript-level verification of every generated statement
- I set up the Nebius Kubernetes cluster, ArgoCD-based GitOps and self-hosted CI runners. Fully automated tag-driven releases run evals and integration tests to validate deployment
- I lead incident command and write postmortem analysis for our production incidents

From 09/2024 - 07/2025 **Expert Software Engineer at Totally Gifts**

- Development of a greenfield project with a new tech stack, from scratch
- NextJS, TailwindCSS, Clerk, Sentry, NestJS, PostgreSQL, Docker, AWS have been used to build the solution
- Event-driven architecture was used to capture gift card events and process them in real-time
- I contributed across the stack, collaborating closely with a cross-functional team of three engineers, a designer, and a product owner
- Management of App Store and Play store releases

From 11/2023 - 09-2024 **Lead Fullstack Software Engineer at Grndhouse**

- Team lead managing a team of 5 backend and frontend developers on grndhouse.com, a platform for on-demand strength training.
- Led the development of the initial MVP after a successful funding round
- Took regular meetings with stakeholders to streamline delivery of video streaming, payment processing, and user management features
- Architected and implemented features using React Native, Node.js, and AWS
- Mentored junior developers, resolved blockers, and ensured timely project delivery

From 12/2021 - 11/2023 **Fullstack Software Engineer at Fertifa**

- Progressed from individual contributor to team lead, managing a team of 5 backend and frontend developers on fertifa.com, a reproductive healthcare provider.
- Defined and developed features in collaboration with the product owner and stakeholders
- Engineered solutions in React Native, Node.js, and AWS

06/2018 – 12/2021 **Senior Software Engineer at KPN Technium B.V.**

- I was senior engineer in team that was developing ETL automation framework and platform in Azure, based on Hadoop & Kafka ecosystem, automated by Terraform
- The rendering framework was built with Python, Flask and SQLAlchemy
- Led story mapping for migration of on-premise data platform services to Azure
- Development of distributed event sourcing application for “real-time” analysis of KPN customer journey where multiple touch points are combined, analysed and visualised
- Scala stack with Akka/Alpakka is used to build microservices and Flink applications
- Utilizing DevOps methodology with fully automated CI/CD, monitoring and alerting

02/2017 – 05/2018 **Software Engineer, Cognitive Implementation at IPsoft B.V.**

- As technical lead of a 3-member agile team I was responsible for development of human-machine interaction on artificial intelligence framework “Amelia”.
- I was mostly busy with hands-on development, code reviewing, mentoring and Scrum rituals.
- Development was mostly in Groovy/Python and in Java
- The technical stack included Kafka, Camel, Mule, Elasticsearch, Spring, Grails, Docker
- Data analysis with Pandas, Jupyter and visualisation with d3.js, and making new features (or refactoring) decisions from the insights

06/2012 – 01/2017 **Senior System Engineer at Liberty Global B.V.**

- As member of Liberty Global OTT streaming solution architecture & engineering team, I was busy with architecture of various system components, developing AWS (micro)services infrastructure, CDN content delivery, Adobe AEM and their integration
- Microservices development in Python

- Through work with high availability services I acquired deep knowledge of web and backend application design, streaming technologies, HA scalable infrastructure, AWS cloud and full stack network protocols
- Data analysis with Jupyter, Wireshark, Conviva, Omniture
- Performing code reviews and overall system troubleshooting

Technical knowledge and other skills

AI and LLM systems

- LLM APIs (OpenAI-compatible: Azure ML, Nebius; Google Gemini), MCP servers and agentic tool use, Pydantic AI and pydantic-evals, prompt design and grounding, LLM-as-judge evaluation, guardrails, LangWatch, PHI de-identification (Deduce)

Languages

- Python, TypeScript, Scala, Java

Frameworks and libraries

- FastAPI, Pydantic, SQLAlchemy/Alembic, PostgreSQL, Node.js/NestJS, Kafka, Spark/Databricks, Flink, pandas
- React, React Native, Next.js, TailwindCSS, Playwright, Jest
- Kubernetes, Docker, ArgoCD, AWS, Azure, Terraform, GitHub Actions, OpenTelemetry, Prometheus/Grafana, Sentry

Languages

- Fluent in English, Dutch (NT2) and Serbian. Intermediate level of German

Education

- 09/2009 – 02/2011 **Masters Course: System Engineering and Radio Communications** – Faculty of Electrical Engineering, University of Belgrade. Specialization: Telecommunications and software engineering
- 10/2002 – 08/2007 **BSc of Electrical Engineering** - Faculty of Electrical Engineering, University of Belgrade. Specialization: Telecommunications and computer science

Certificates

- Acclaim badges: <https://www.credly.com/users/dejan-petkovic/badges>
- Coursera certificates: <https://coursera.org/share/32a42acb44082359b7c10bfea2f765ee>

Projects

- BrAln: <https://github.com/dexpetkovic/brAln>
 - o WhatsApp-based AI assistant built with NestJS and Google Gemini. Handles WhatsApp webhook events and generates contextual replies, with an MCP server that lets the model store, update and retrieve memories.
- Biller: <https://biller.elands.studio>
 - o Professional invoicing for freelancers. Create, send, and track invoices in seconds. One-time purchase, no subscriptions.
- Bouwen: <https://bouwen.elands.studio>
 - o Renovation project communication for the Dutch market. Print a QR code for your window and let neighbours follow your renovation in real time.
- 2e-woning.nl: <https://2e-woning.nl>
 - o Box 3 tax calculator for a second home and other investments. Built for the Dutch market.